

IET DAVV INDORE
2nd year IT Sec A and Sec B
TR3C4-Digital Electronics class test -2

Time :70min

Max Marks: 20

Note: solve any four questions. All questions carry equal marks. Make suitable assumptions whenever required.

- ✓ Q.1 Explain combinational circuit. Draw and explain the working of adder subtractor circuit. 5
- Q.2 Design a 4-line to 2-line priority encoder. Include an output E to indicate that at least one input is a 1. 5
- ✓ Q.3 Design BCD- to -excess-3 code converter and explain clearly when input bit is 1000 then what will be the output of this code converter. 5
- ✓ Q.4 Explain magnitude comparator with the help of logic diagram. 5
- Q.5 Get simplified expression of $Y = F(A, B, C, D) = \sum m(1, 2, 8, 9, 10, 12, 13, 14)$ using Quine-McClusky method. 5